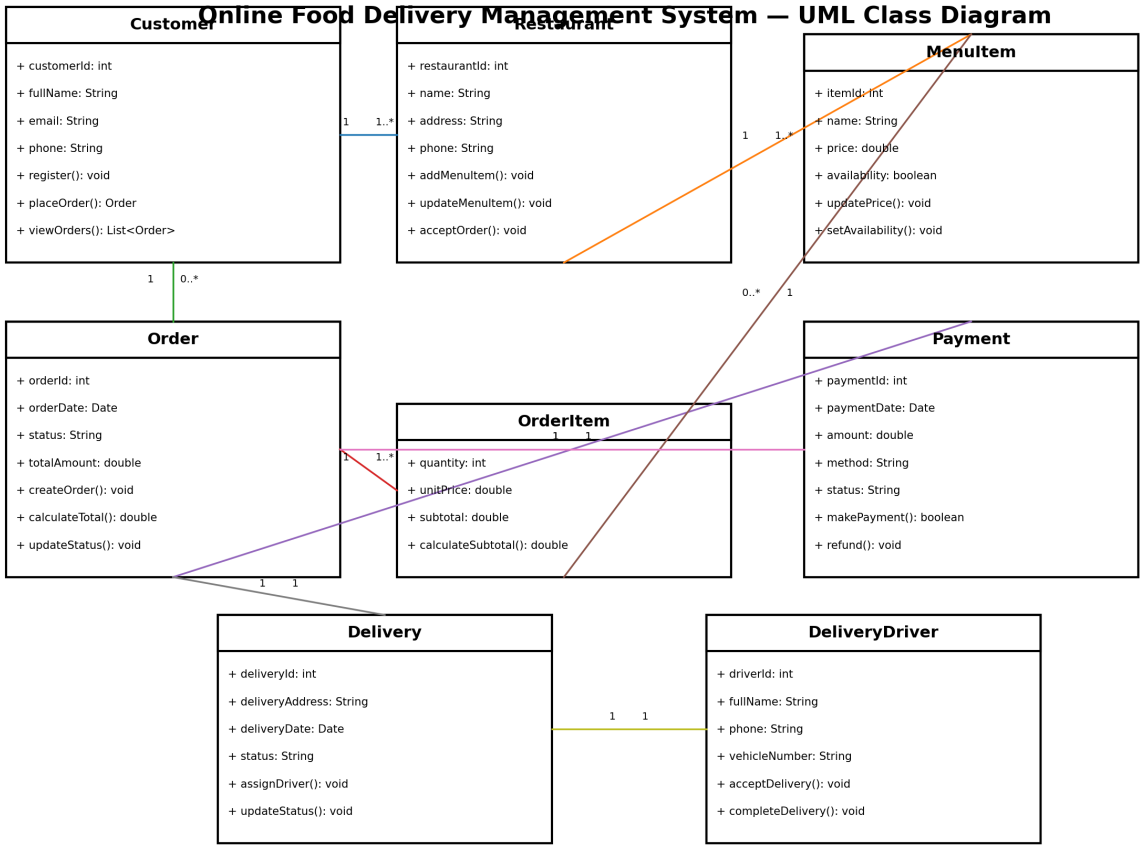
Online Food Delivery Management System

Report 2: UML Class Diagram Report

1. Diagram Overview

The UML class diagram models the main entities of the online food delivery system, their attributes, operations, relationships, and multiplicities.



2. Classes, Attributes, and Methods

Class	Attributes	Methods
Customer	customerId, fullName, email, phone	register(), placeOrder(), viewOrders()
Restaurant	restaurantId, name, address, phone	addMenuItem(), updateMenuItem(), acceptOrder()
MenuItem	itemId, name, price, availability	updatePrice(), setAvailability()
Order	orderId, orderDate, status, totalAmount	createOrder(), calculateTotal(), updateStatus()
OrderItem	quantity, unitPrice, subtotal	calculateSubtotal()
Payment	paymentId, paymentDate, amount, method, status	makePayment(), refund()
Delivery	deliveryId, deliveryAddress, deliveryDate, status	assignDriver(), updateStatus()
DeliveryDriver	driverId, fullName, phone, vehicleNumber	acceptDelivery(), completeDelivery()

3. Relationships and Multiplicities

Relationship	Multiplicity	Explanation
Customer — Restaurant	1 to 1..*	A customer can interact with one or more restaurants; a restaurant can serve many custom
Restaurant — MenuItem	1 to 1..*	A restaurant manages one or more menu items.

Relationship	Multiplicity	Explanation
Customer — Order	1 to 0..*	A customer may place zero or many orders; each order belongs to one customer.
Order — OrderItem	1 to 1..*	Each order contains one or more order items.
OrderItem — MenuItem	0..* to 1	A menu item can appear in many order items; each order item refers to one menu item.
Order — Payment	1 to 1	Each order has one payment record in this model.
Order — Delivery	1 to 1	Each order has one delivery record.
Delivery — DeliveryDriver	1 to 1	Each delivery is assigned to one driver in this model.

4. Relationship Explanation

The Customer places Orders. Each Order is composed of OrderItems, and each OrderItem refers to a MenuItem supplied by a Restaurant. An Order has a Payment record and a Delivery record. Each Delivery is assigned to a DeliveryDriver. The multiplicities communicate how many instances of each class can participate in a relationship.